

Label 3-D Medical Image Using Medical Image Labeler

This example shows how to label 3-D image data using the Medical Image Labeler app. The Medical Image Labeler app provides manual, semi-automated, and automated tools for labeling 3-D medical image data. This example segments a chest CT volume to label a lung tumor region.

Try This Example

 Copy Command

Download Data to Label

This example labels chest CT data from a subset of the Medical Segmentation Decathlon data set [1]. The size of the subset of data is approximately 76 MB. Run this code to download the MedicalVolumeNIIFTIData.zip file from the MathWorks® website, then unzip the file.

```
zipFile = matlab.internal.examples.downloadSupportFile("medical", "MedicalVolumeNIIFTIData.zip");
filepath = fileparts(zipFile);
unzip(zipFile, filepath)
```

 Get ▾

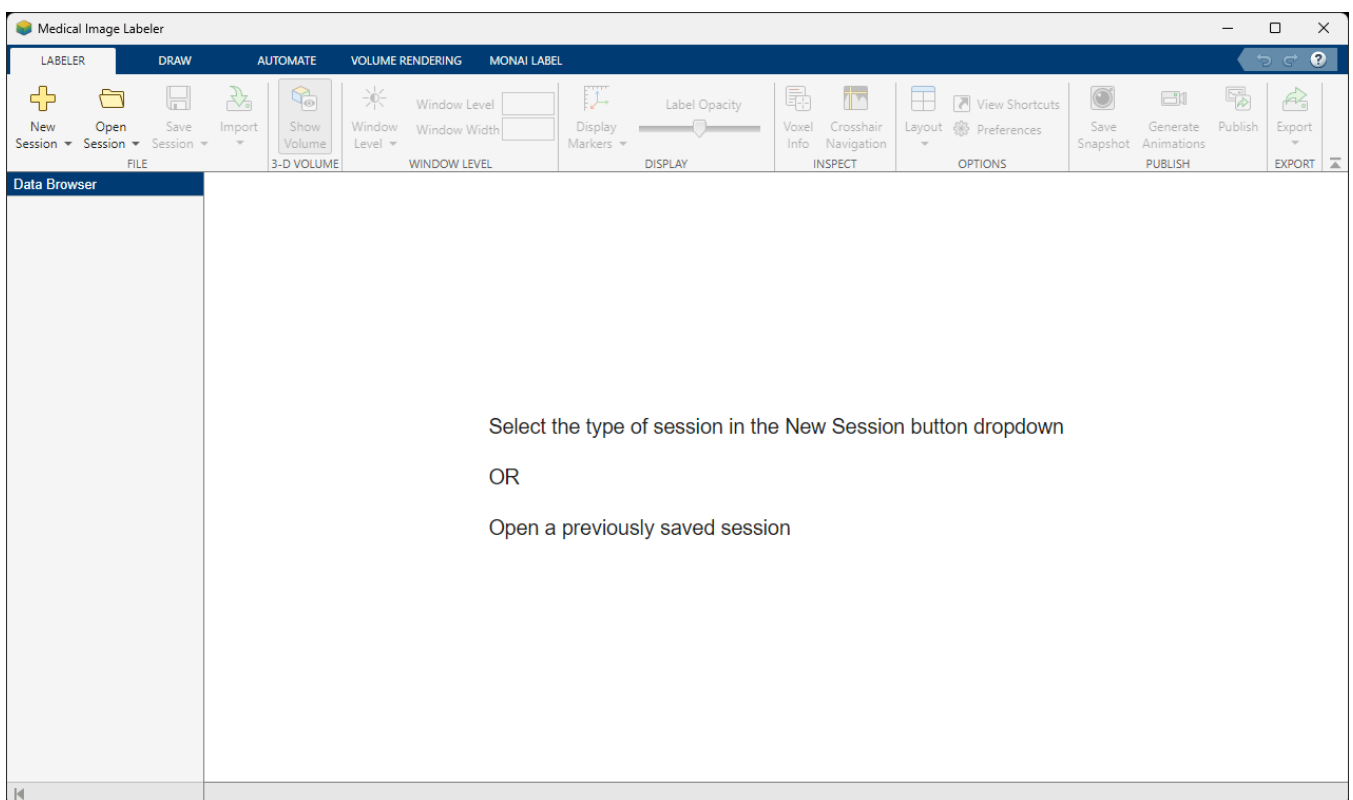
The dataFolder folder contains the downloaded and unzipped data.

```
dataFolder = fullfile(filepath, "MedicalVolumeNIIFTIData");
```

 Get ▾

Open the Medical Image Labeler

Open the **Medical Image Labeler** app from the **Apps** tab on the MATLAB® toolstrip, under **Image Processing and Computer Vision**. You can also load the app by using the `medicalImageLabeler` command.



Create New Volume Labeling Session

To start a new 3-D labeling session, on the app toolstrip, click **New Session** and select **New Volume session (3-D)**. In the **Create a new session folder** dialog box, specify a location in which to save the new session folder by entering a path or selecting **Browse** and navigating to your desired location. In the **New Session Folder** dialog box, specify a name for the folder for this labeling session. Then, select **Create Session**.

Load Image Data into Medical Image Labeler

To load an image into the **Medical Image Labeler** app, on the app toolstrip, click **Import Data**. Then, under **Data**, select **From File**. Browse to the location where you downloaded the data. You can find the data location by checking the `dataFolder` value in the workspace. Once you browse to the download location, press **Shift** and select the files `lung_027.nii.gz` and `lung_043.nii.gz`. Click **Open** to import the files. For a **Volume Session**, the imported data file can be a single DICOM or NIFTI file containing a 3-D image volume, or a directory containing multiple DICOM files corresponding to a single image volume.

The name of the imported image is visible in the **Data Browser** pane. The no labels symbol  next to the file name indicates that the volume does not contain any pixel labels. You can import multiple 3-D image files into a **Volume Session**. All files imported into a single app session must label the same regions of interest, such as tumor or lung, and are exported together as one `groundTruthMedical` object.

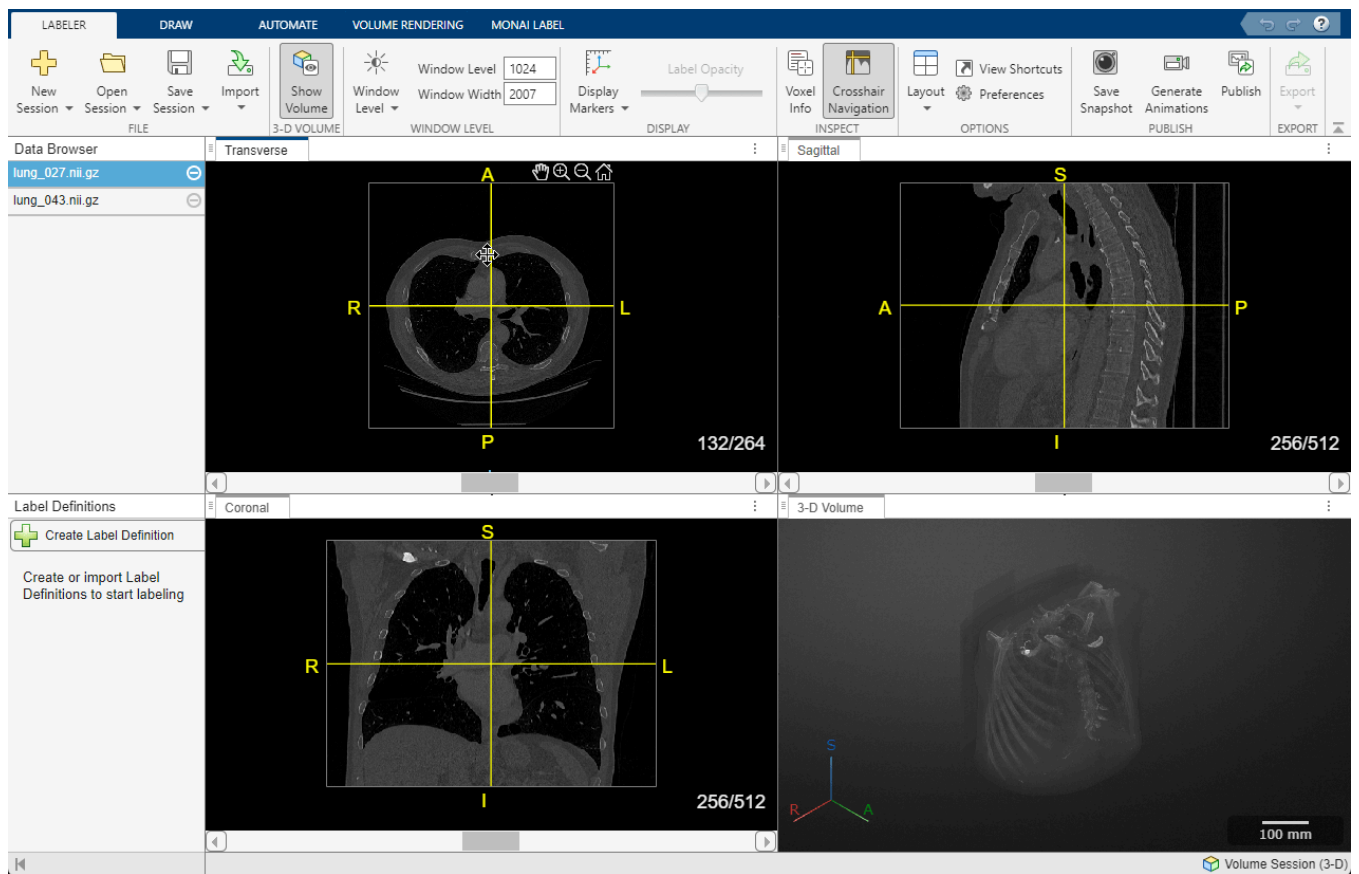


Explore the Image Volume

The **Medical Image Labeler** app displays a 3-D rendering of the scan in the **3-D Volume** pane, and displays anatomical slice planes in the **Transverse**, **Sagittal**, and **Coronal** panes. You can toggle the visibility of the volume display using the **Show Volume** button on the **Labeler** tab of the app toolbar.

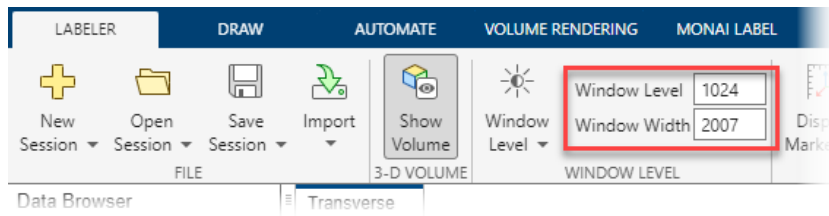
By default, the app displays the center slice in each slice plane. You can change which slice is displayed by using the scroll bar at the bottom of a slice pane, or you can click the pane and then press the left and right arrow keys. The app displays the current slice number out of the total number of slices, such as 132/264, for each slice pane. The app also displays anatomical display markers indicating the anterior (A), posterior (P), left (L), right (R), superior (S), and inferior (I) directions. You can zoom in on the current slice pane using the mouse scroll wheel or the zoom controls that appear when you pause on the slice pane.

You can also navigate between slices in the three planes using crosshair navigation. To turn on the crosshair indicators, in the **Labeler** tab of the app toolbar, select **Crosshair Navigation**. The indicators show the relative positions of the slices in the other 2-D views. To navigate slices, pause on one of the crosshair axes until the cursor changes to the fleur shape, , and then click and drag to a new position. The other slice views update automatically. To hide the crosshairs, in the app toolbar, clear **Crosshair Navigation**.



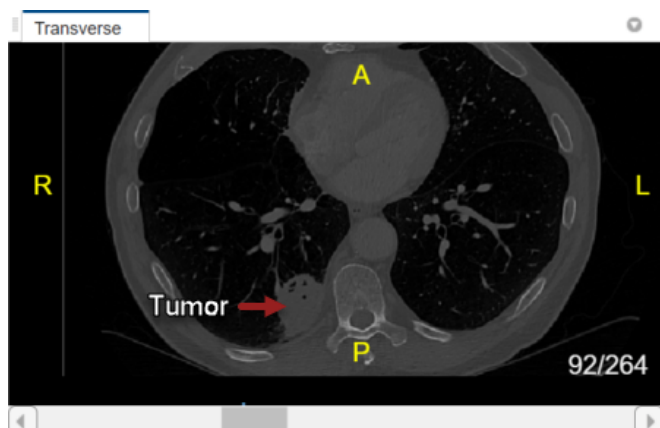
By default, the app displays the scan using the **Radiological** display convention, with the left side of the patient on the right side of the image. To display the left side of the patient on the left side of the image, on the app toolbar, click **Preferences**, and in the dialog box that opens, change the **Display Convention** to **Neurological**.

To change the brightness or contrast of the display, adjust the display window. Changing the display window does not modify the underlying image data. To update the display window, you can directly edit the values in the **Window Level** and **Window Width** boxes in the app toolbar.



Alternatively, adjust the window using the **Window Level** tool. First, select  in the app toolbar. Then, click and hold in any of the slice panes, and drag up and down to increase and decrease the brightness, respectively, or left and right to increase and decrease the contrast. The **Window Level** and **Window Width** values in the toolbar automatically update as you drag. Click **Clear** to deactivate the tool. To reset the brightness and contrast, click **Window Level** and select **Reset**. You can swap the mouse actions, such that dragging vertically changes the contrast and dragging horizontally changes the brightness, in the **Preferences** dialog box.

With `lung_027.nii.gz` selected in the **Data Browser** pane, explore the chest volume to identify the tumor region you want to label. The tumor is visible between slices 79 and 105 in the **Transverse** slice pane.



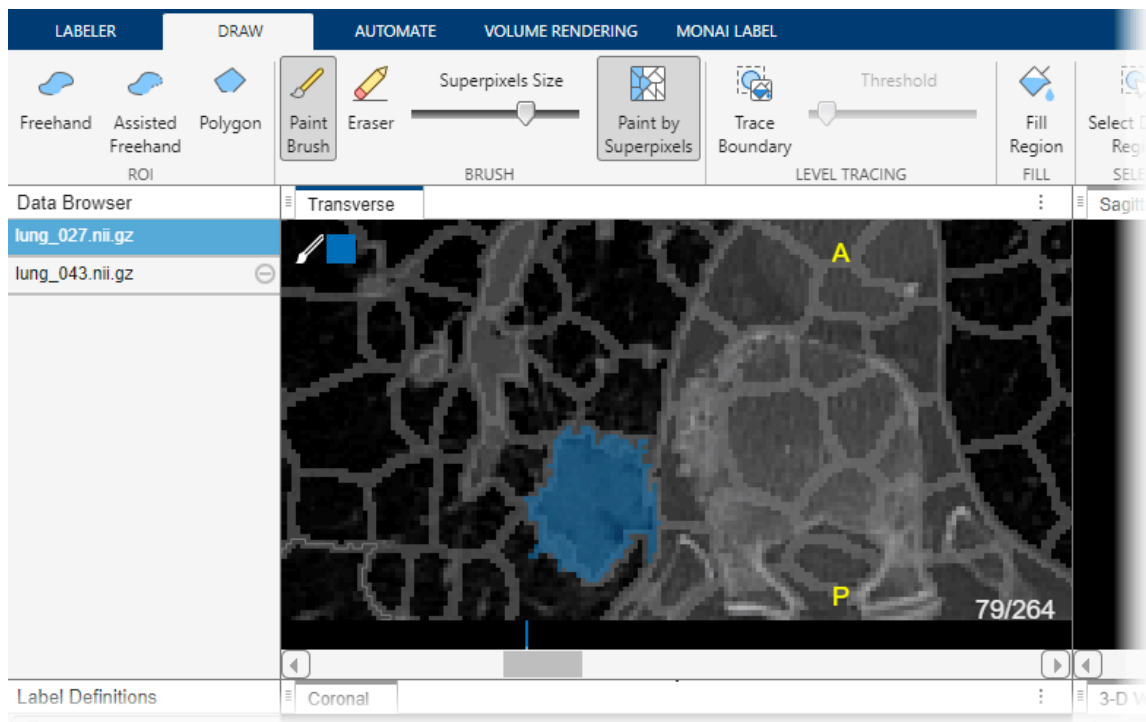
Create Label Definitions

A *label definition* specifies the name, color, and numeric index of a label. In the **Label Definitions** pane, select **Create Label Definition** to create a label with the default name Labe11. To change the name of the label, double-click on the label and type in a new name. The label name must be a valid MATLAB® variable name with no spaces. For this example, specify the name of the label as tumor. To change the default color associated with the label, double-click on the colored square in the label identifier and select a color from the Color dialog box.



Use Drawing Tools to Label Regions in Image

Click on the tumor label in the **Label Definitions** pane to assign pixels to the tumor label. You can use the tools on the **Draw** tab in the app toolstrip to define the region. To learn more about each of the available drawing tools, see *Get Started with Medical Image Labeler*. This example uses the **Paint Brush** tool, with **Paint by Superpixels** enabled, to label the tumor. You can draw labels in the **Transverse**, **Sagittal**, or **Coronal** slice panes. For this example, label slice 79 in the **Transverse** pane. When you add the label, the app adds a bar above the slider, using the color associated with the label, to indicate which slices you have labeled.

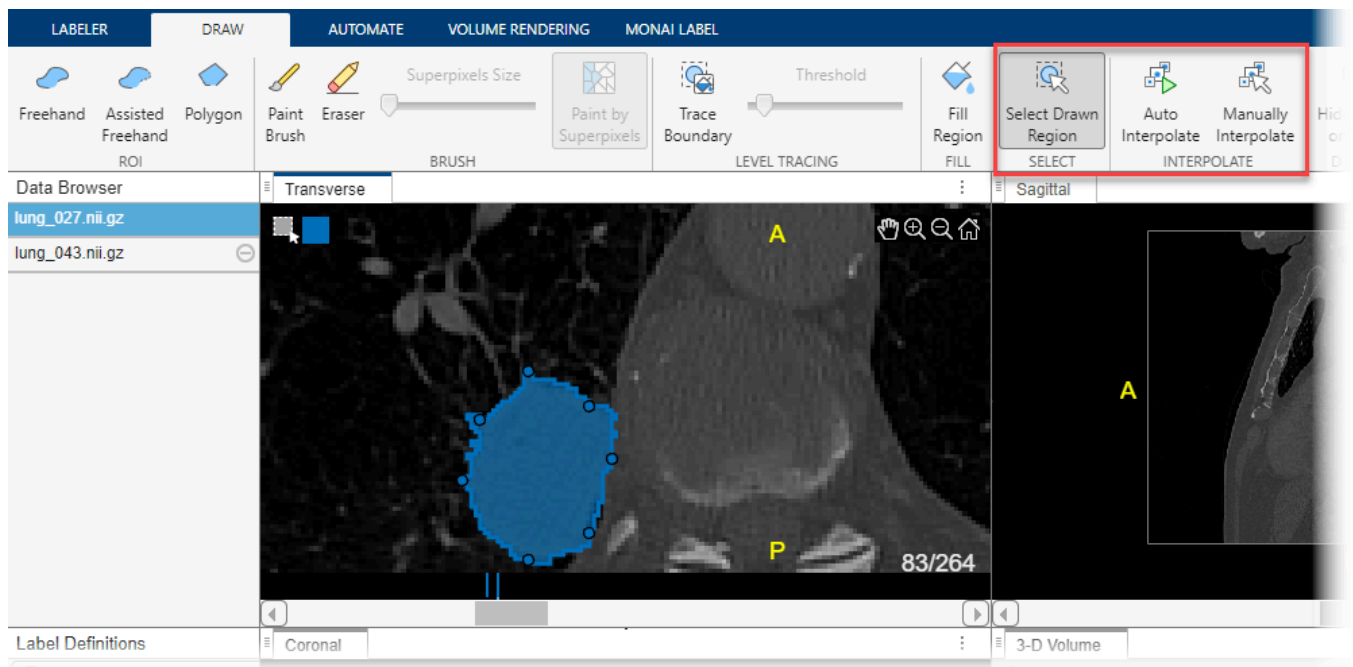


In addition to the drawing tools, you can add or refine label regions using the tools in the **Automate** tab of the app toolstrip. The app provides slice-based automation algorithms including **Active Contours**, **Adaptive Threshold**, **Dilate**, and **Erode**, as well as volume-based algorithms including **Filter and Threshold**, **Smooth Edges**, and **Otsu's Threshold**. To apply a slice-based algorithm, click **Algorithm Parameters** to adjust settings if applicable, select an option from the **Slice Direction** list, and click **Run**. You can also specify a custom range of slices to process by specifying a **Start** slice and an **End** slice.

Use Interpolation to Speed Up Labeling

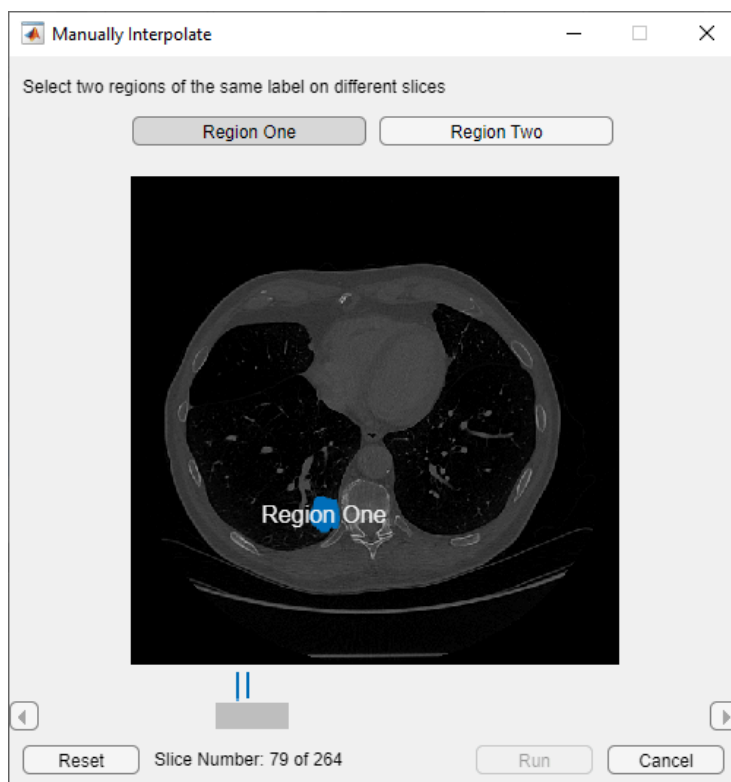
You can move through the image volume and draw labels slice-by-slice. The **Medical Image Labeler** app also provides interpolation tools in the **Draw** tab that can help with labeling an object between slices.

To use interpolation, you must first manually label a region on two slices. You have already labeled the tumor in slice 79. Use the same process to label the tumor in slice 83. If the **Auto Interpolate** button is not active, make sure the labeled region is selected by clicking **Select Drawn Region** in the app toolstrip and selecting the labeled region.



Click **Auto Interpolate**. The app automatically labels the tumor in the intermediate slices. The app adds bars above the slider to indicate all the slices that have labeled pixels, which now appears as a solid bar from slice 79 to slice 83.

Alternatively, after labeling an ROI on two slices, you can click **Manually Interpolate**. With this option, the app opens the Manually Interpolate dialog box. Select the two regions between which you want to interpolate, **Region One** and **Region Two**. To select the first region, use the slider at the bottom of the dialog box to navigate to slice 79, and then click inside the labeled tumor. To select the second region, click **Region Two**, navigate to slice 83, and click inside the labeled tumor. After selecting both regions, click **Run** to interpolate the label in the intermediate slices.



After using interpolation, check the individual slices to see if the interpolation created satisfactory labels. You can manually correct labels using the **Paint Brush** and **Eraser** tools. Alternatively, you can refine the labels using one of the tools in the **Automate** tab. For example, you can use **Active Contours** to grow an ROI in a slice where it does not fill the full area of the tumor. You can also undo the interpolation and try interpolating across fewer slices to improve results.

Visualize and Modify Labels

As you draw labels, you can view them as an overlay in the **3-D Volume** pane. In the warning message at the top of the **3-D Volume** pane, click **Update**. Note that **Update** is only available after you update the label data in a slice pane. To adjust the opacity of all labels in the session, in the **Labeler** tab of the app toolbar, adjust the **Label Opacity** slider. To toggle the visibility of an individual label, in the **Label Definitions** pane, click the eye symbol  next to the label name.



To refine drawn labels, you can remove label data from individual pixels, from individual slices within an image, from an entire image, or from the whole labeling session.

- Remove labels from individual pixels — In the **Draw** tab, use the **Eraser** tool.
- Remove labels from one connected region in a slice — Click **Select Drawn Region** and, in the slice pane, select the region from which you want to remove labels. You can also select the label to remove by right-clicking on the label within the slice plane and selecting **Select Drawn Region**. Press **Delete**, or right-click and select **Delete**, to remove labels from the region.
- Remove all labels from a slice — Right-click anywhere on the slice and select **Select All**. Press **Delete**, or right-click and select **Delete**, to remove all labels from the slice.
- Remove all labels from an image volume — Right-click the file name of the image volume in the **Data Browser** and select **Remove Labels**. Removing labels from an image volume removes all pixel labels for all label definitions within the file.
- Remove a label from all images within the app session — In the **Label Definitions** pane, right-click the label name and select **Delete**. This deletes the label from the `groundTruthMedical` object in the session folder, and removes all pixel labels for the deleted label name in all images in the session.

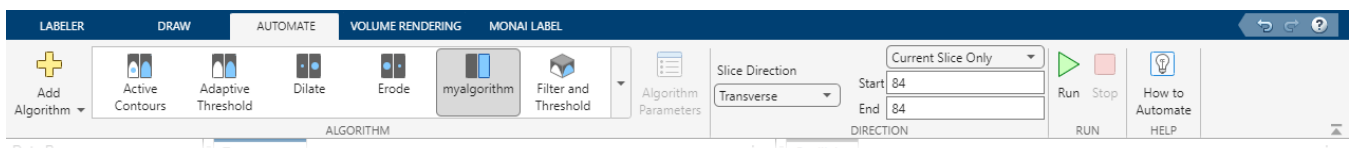
Apply Custom Automation Algorithm

Optionally, you can add a custom automation algorithm to use in the app. See [Automate Labeling in Medical Image Labeler](#) for an example that applies a custom automation algorithm for 2-D labeling.

To add an algorithm, on the **Automate** tab, click **Add Algorithm**. You can specify a **Slice-Based** or a **Volume-Based** algorithm, where slice-based algorithms operate slice-by-slice along one direction, and volume-based algorithms operate on the volume in 3-D. Import an existing algorithm by selecting **From File**, or create a new algorithm by selecting **New > Function Template**. The template opens in the MATLAB editor. Replace the sample code in the template with code that you want to use. Your function must accept two arguments: each slice as a separate image, and a mask. Your function must also return a mask image. When you are done editing the template, save the file. The **Medical Image Labeler** app automatically creates a button in the **Automate** tab toolbar for your function.

To test a volume-based function, select your algorithm and click **Run**. The algorithm processes the volume. The slices panes update to show the new labels.

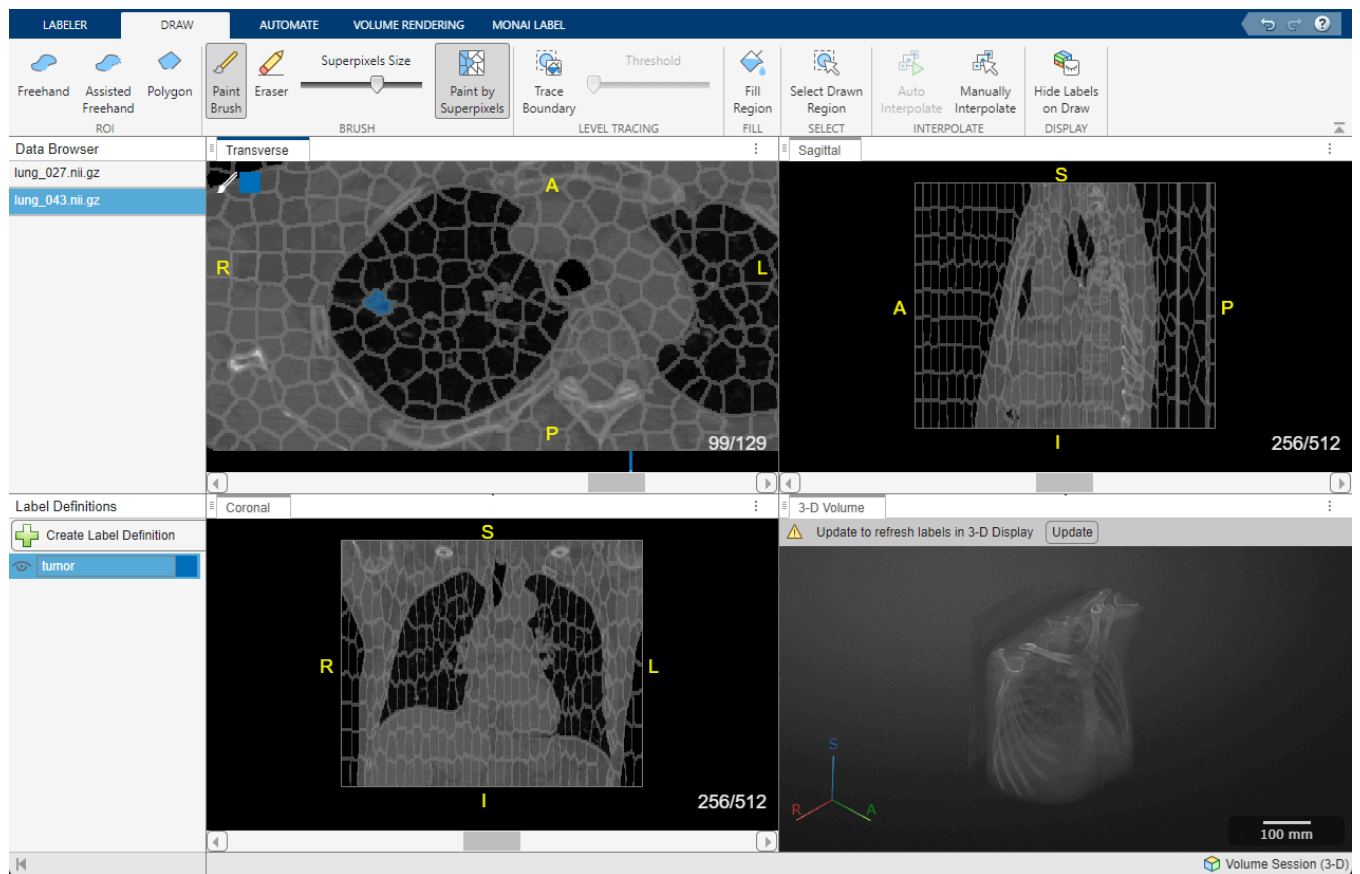
To test a slice-based algorithm, select your algorithm, choose the **Slice Direction** to apply the algorithm in, and click **Run**. By default, the app applies the function to only the current slice. After testing your function on a single slice, you can run it on all of the slices, or a subset of the slices, in the selected direction. You can run it from the current slice to the end (the highest numbered slice) or from the current slice back to the beginning (slice 1). You can also specify a range of slices by specifying the starting slice and the ending slice.



Label Next Image Volume

For this example, the labels for `lung_027.nii.gz` are complete when you have labeled each slice of the tumor. **Medical Image Labeler** automatically assigns a value of 0 to unlabeled pixels in the label images saved to the session folder, so you do not need to label the background manually.

To move to the next volume, select `lung_043.nii.gz` in the **Data Browser** pane. Repeat the labeling process to draw labels on the tumor region in this volume.



Export Ground Truth Data

The **Medical Image Labeler** app automatically saves a `groundTruthMedical` object as a MAT file in the session folder. You can also export a `groundTruthMedical` object, saved as a MAT file, to an alternate file location from the app. On the **Labeler** tab, click **Export** and, under **Ground Truth**, select **To File**.

You can load the exported MAT file into the MATLAB workspace using the `load` function. The properties of the `groundTruthMedical` object, `gTruthMed`, contain information about the image data source, label definitions, and location of the saved label images. Display information about the object and each of its properties using these commands.

- `gTruthMed` — Display the properties of the `groundTruthMedical` object.
- `gTruthMed.DataSource` — Location of the source of the unlabeled medical volumes.
- `gTruthMed.LabelDefinitions` — Table of information about label definitions.
- `gTruthMed.LabelData` — Locations of the saved, labeled images.

References

[1] Medical Segmentation Decathlon. "Lung." Tasks. Accessed May 10, 2018. <http://medicaldecathlon.com/>.

The Lung data set is provided by the Medical Segmentation Decathlon under the CC-BY-SA 4.0 license. All warranties and representations are disclaimed. See the license for details. This example uses a subset of the original data set consisting of two CT volumes. The labels shown in this example were created for illustration purposes and have not been verified by a clinician.

See Also

Medical Image Labeler | `groundTruthMedical`

Related Topics

- Get Started with Medical Image Labeler
- Label 2-D Ultrasound Series Using Medical Image Labeler
- Automate Labeling in Medical Image Labeler